

Project Initialization and Planning Phase

Date	15-07-2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposed project, Rising Waters – Flood Prediction System Using Machine Learning, aims to provide an intelligent flood prediction solution using historical disaster and environmental data. The system leverages Machine Learning algorithms to analyze factors such as rainfall, elevation, slope, geographical location, and disaster statistics to predict the likelihood of flooding. This solution helps authorities and citizens take preventive measures, reducing the impact of flood-related disasters. The project focuses on improving prediction accuracy, supporting early warning systems, and enhancing disaster preparedness through a user-friendly web application.

Project Overview	
Objective	The primary objective is to develop an intelligent flood prediction system using Machine Learning that accurately predicts flood occurrences based on environmental and disaster-related parameters, enabling early warning and effective disaster management.
Scope	The project includes data preprocessing, machine learning model development, prediction of flood risk, visualization of results, and deployment through a Flask-based web application for real-time user interaction.
Problem Statement	
Description	Traditional flood forecasting methods are often slow, less accurate, and unable to provide timely warnings. The absence of intelligent prediction systems increases the risk of property damage, economic loss, and threats to human life during flood events.
Impact	Implementing an intelligent flood prediction system will improve disaster preparedness, provide faster and more accurate flood predictions, reduce economic losses, support emergency response teams, and help protect lives and property.
Proposed Solution	
Approach	Develop a Machine Learning-based flood prediction system that analyzes environmental and disaster-related data to predict flood occurrence. The system uses a trained model integrated with a Flask web application to provide instant predictions and visual analytics.
Key Features	- Machine Learning-based flood prediction model. - Real-time flood risk prediction through a web interface. - Interactive dashboard with graphical visualization using Chart.js. - User-friendly interface for entering environmental parameters. - Fast prediction results with flood risk percentage. - Scalable architecture for future cloud deployment and live weather API integration.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
	Computing Resources	Intel Core i5/i7 Processor (or equivalent)
	Memory	8 GB RAM or higher
	Storage	1 TB HDD / 512 GB SSD
Software		
	Framework	Flask
	Libraries	Scikit-learn, Pandas, NumPy, Joblib, Matplotlib, Chart.js
	Development Environment	Visual Studio Code, Jupyter Notebook
Data		
Data	Source, Size, Format	Kaggle Flood Dataset, CSV Format